

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Fifth Semester B.E. (CCE) (Credit System) Degree Examinations**

January 2023

20CC504 - NETWORK ENGINEERING

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

Unit – I		Marks	BT*	CO*	PO*
1.	a) What are the strategies to enhance the available capacity and lower latencies in a data network? What is the complexity involved in implementing each of them?	10	L*2	1	1
	b) Discuss TCP Slow Start. Discuss its impact on the networks with high bandwidth and throughput.	10	L4	2	1
2.	a) With a neat timing diagram explain the TCP Three-way Handshake mechanism. What is the purpose of using multiple random variables and why they are random?	10	L4	1	1
	b) With 16KB each congestion window and receiver window size find the data transfer rate on a TCP connection with 100msec RTT. What is the utilization of the bandwidth in this scenario? If sender has 10Mbps bandwidth available and receiver has 100 Mbps bandwidth available calculate the minimum congestion window that could saturate the bandwidth between the sender and receiver.	5	L3	2	1
	c) Briefly explain Proportional Rate Reduction for TCP and Multiplicative Decrease and Additive Increase concepts.	5	L2	2	1
3.	a) Define latency and bandwidth. Distinguish between latency and propagation delay. List and explain the 4 major components of Latency. What is the hard limit of the propagation of an electrical signal?	10	L2	1	1
	b) What is the implication of window scaling on the network throughput? List three major implementation default values for TCP window scaling introduced. With a suitable example distinguish the data delivery efficiency of a TCP session with any two different scaling values.	10	L3	2	1
Unit – II					
4.	a) Define congestion avoidance. With a suitable example explain how active queue management is carried out using DECbit in a TCP session.	10	L2	3	1
	b) What is ECN? Draw the respective protocol headers clearly specifying the bits used in ECN implementation. Explain the interpretation of the combination of these bits when TCP is enabled with ECN. Explain the process of ECN Negotiation.	10	L3	4	1
5.	a) Differentiate between congestion control and congestion avoidance. Discuss major three components of the RED algorithm.	10	L2	3	1
	b) Explain the working of ECN+. What are the enhancements provided by ECN+algorithm over ECN?	10	L2	4	1
6.	a) Discuss the motivation and main contributions in Adaptive RED. Distinguish between Adaptive RED and Self Configuring RED.	10	L4	3	1
	b) Explain the working of ECN+/wait. What are the enhancements provided by ECN+/TryOnce algorithm over ECN+/wait?	10	L2	4	1

P.T.O.

Unit – III

7.	a)	Explain the Controlled Delay AQM algorithm.	10	L2	5	1
	b)	Discuss the calculation of drop probability of PI and PIE algorithms.	10	L2	5	1
8.	a)	Differentiate decision-making logic to decide whether the incoming packet should be enqueued or dropped with PI and PIE algorithms. How calculation of instantaneous queue delay is done with PIE algorithm?	10	L2	5	1
	b)	Explain the DCN architecture and its components with a suitable neat diagram.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
